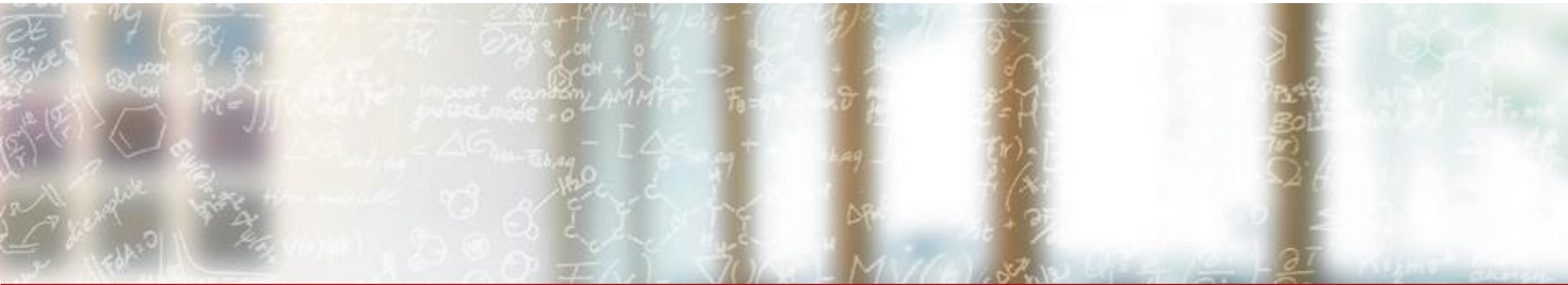




CSCS

Centro Svizzero di Calcolo Scientifico
Swiss National Supercomputing Centre

ETH zürich



HPC Console

CSCS User Lab Day 2026

August 28th, 2026

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Luzern, Switzerland

Agenda

- Introduction and motivation
- HPC Console and key workflows
- Future work and summary
- Demo
- Audience feedback

Motivation

- HPC systems are traditionally accessed through **command-line tools like SSH**
- Interactive environments like **JupyterHub** can make some workflows more accessible
- These tools provide **flexibility and control**, but require technical expertise
- For **new users and users outside computer science**, the learning curve can be steep
- Many everyday HPC tasks **don't require full command-line access**
- Users often just need to **check system status, submit or monitor jobs, and view results**

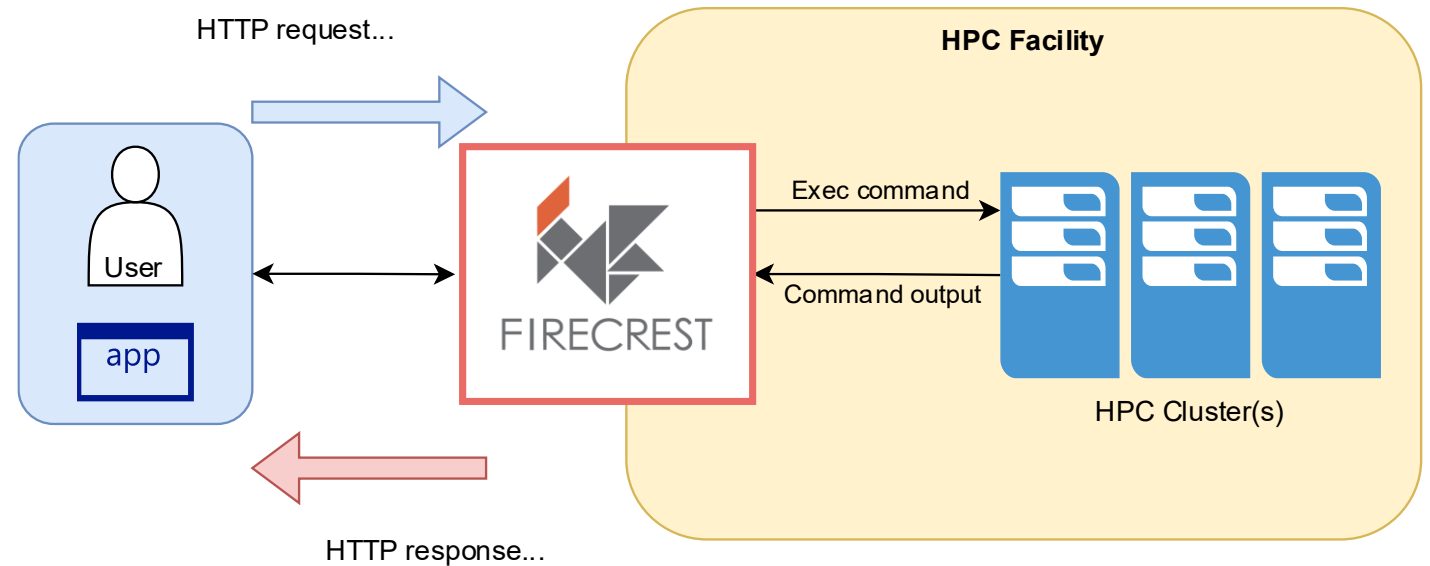
What if the most common HPC tasks were available directly through a web interface?

The HPC Console

- Built on top of **FirecREST** APIs
- Provides a unified interface for accessing HPC resources
- Complements existing tools like SSH and [JupyterHub](#)
- Open-source as **firecrest-ui**, deployed at CSCS as **HPC Console**



FirecREST API



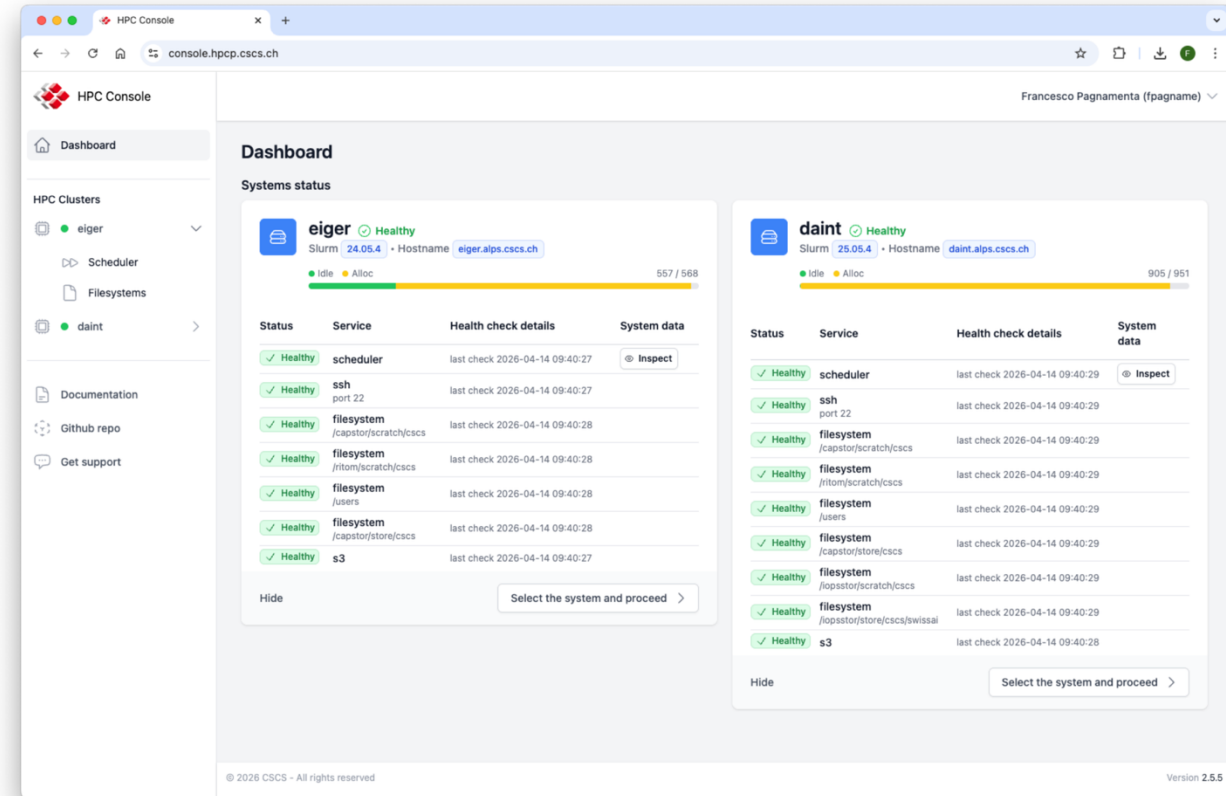
Core Functionality

- Dashboard
 - Inspect cluster health and system availability
- Cluster – Scheduler
 - List batch jobs
 - Configure and submit batch jobs
 - Monitor running jobs and inspect outputs
- Cluster – Filesystem
 - Manage data and perform file operations
 - Preview small files
 - Large data upload/download

Dashboard: cluster health overview

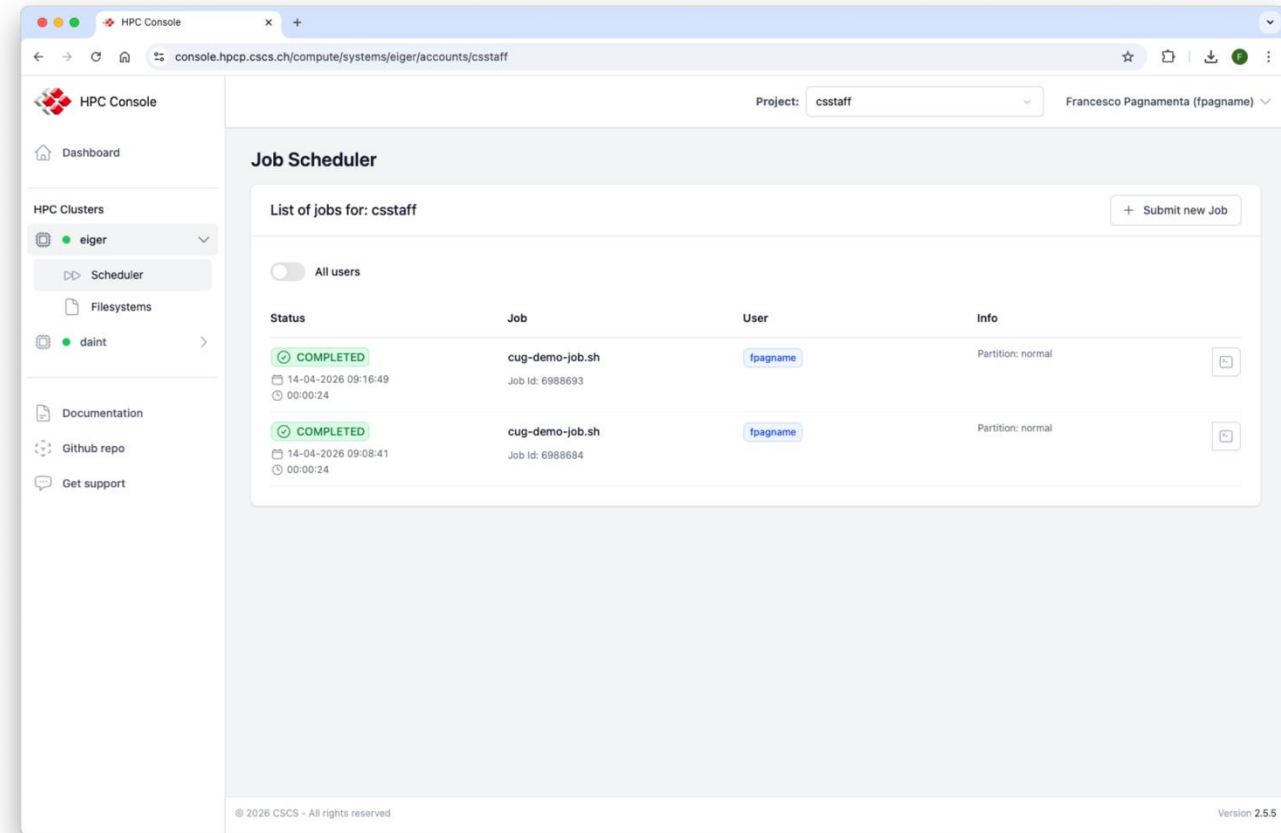
- Dashboard:
 - ☐ Inspect cluster health and system availability
- Cluster – Scheduler:
 - ☐ List jobs
 - ☐ Submit and configure new workloads
 - ☐ Monitor running jobs and inspect outputs
- Cluster – Filesystem
 - ☐ Manage data and perform file operations
 - ☐ Preview small files
 - ☐ Large data upload

✓ The dashboard provides a real-time overview of cluster health



Scheduler: job listing and filtering

- Dashboard:
 - Inspect cluster health and system availability
- Cluster – Scheduler:
 - List jobs
 - Submit and configure new workloads
 - Monitor running jobs and inspect outputs
- Cluster – Filesystem
 - Manage data and perform file operations
 - Preview small files
 - Large data upload

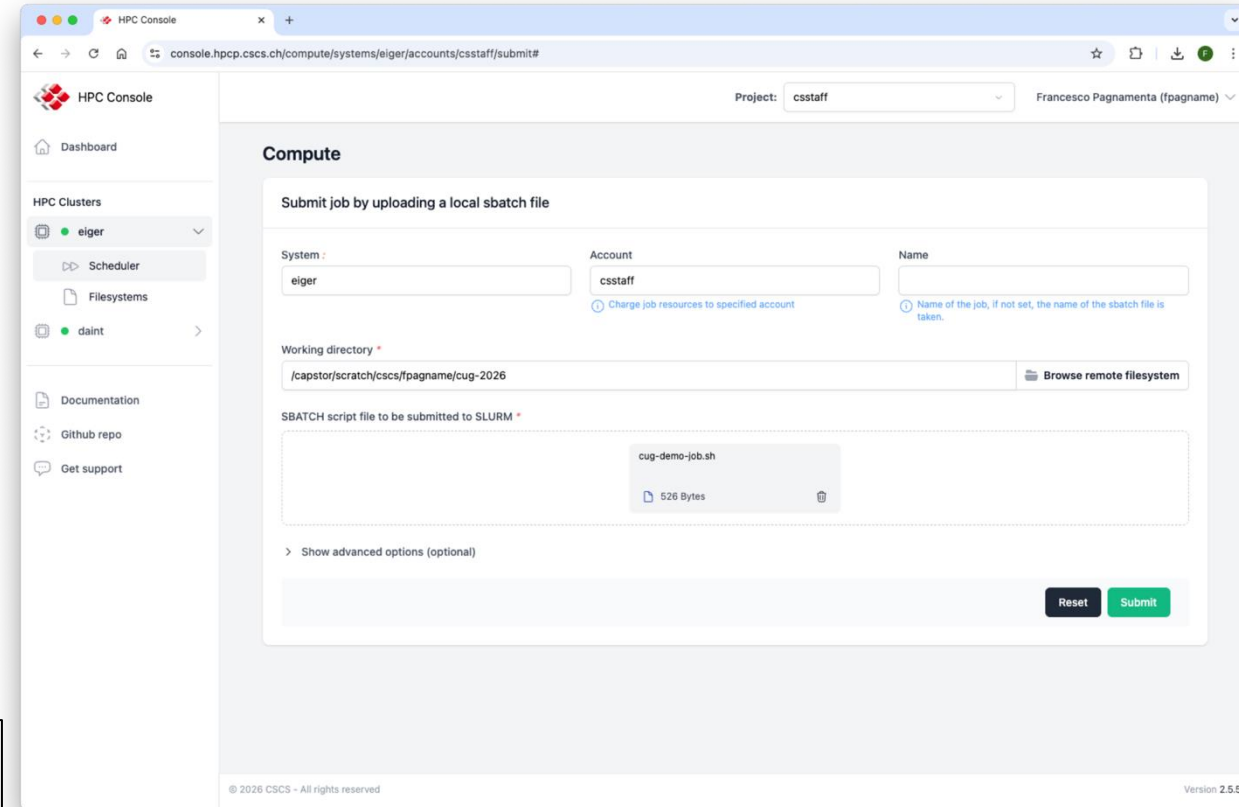


✓ A flag allows listing all jobs associated with the same account

Job submission: configuration and execution

- Dashboard:
 - Inspect cluster health and system availability
- Cluster – Scheduler:
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 - Submit and configure new jobs
 - Monitor running jobs and inspect outputs
- Cluster – Filesystem
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 - Preview small files
 - Large data upload

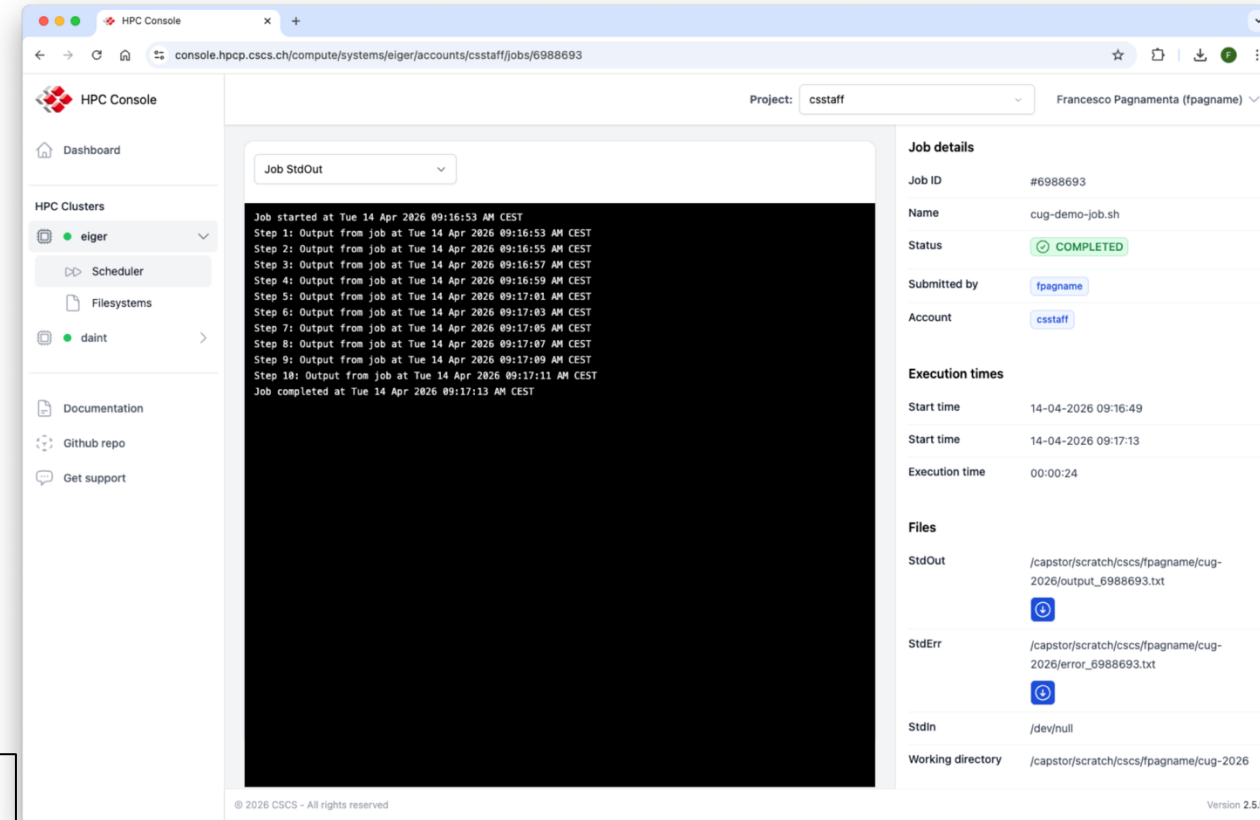
- ✓ Users can detect degraded services before submitting jobs
- ✓ The system prevents job execution on unhealthy clusters
- ✓ Jobs can be submitted and configured directly from the UI



Monitoring: job inspection and logs

- Dashboard:
 - Inspect cluster health and system availability
- Cluster – Scheduler:
 - List jobs
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- Cluster – Filesystem
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 - Preview small files
 - Large data upload

- ✓ Users can inspect job scripts, logs, and execution status
- ✓ Each job has a shareable URL for collaboration



Filesystem: data management

■ Dashboard:

- ☐ Inspect cluster health and system availability

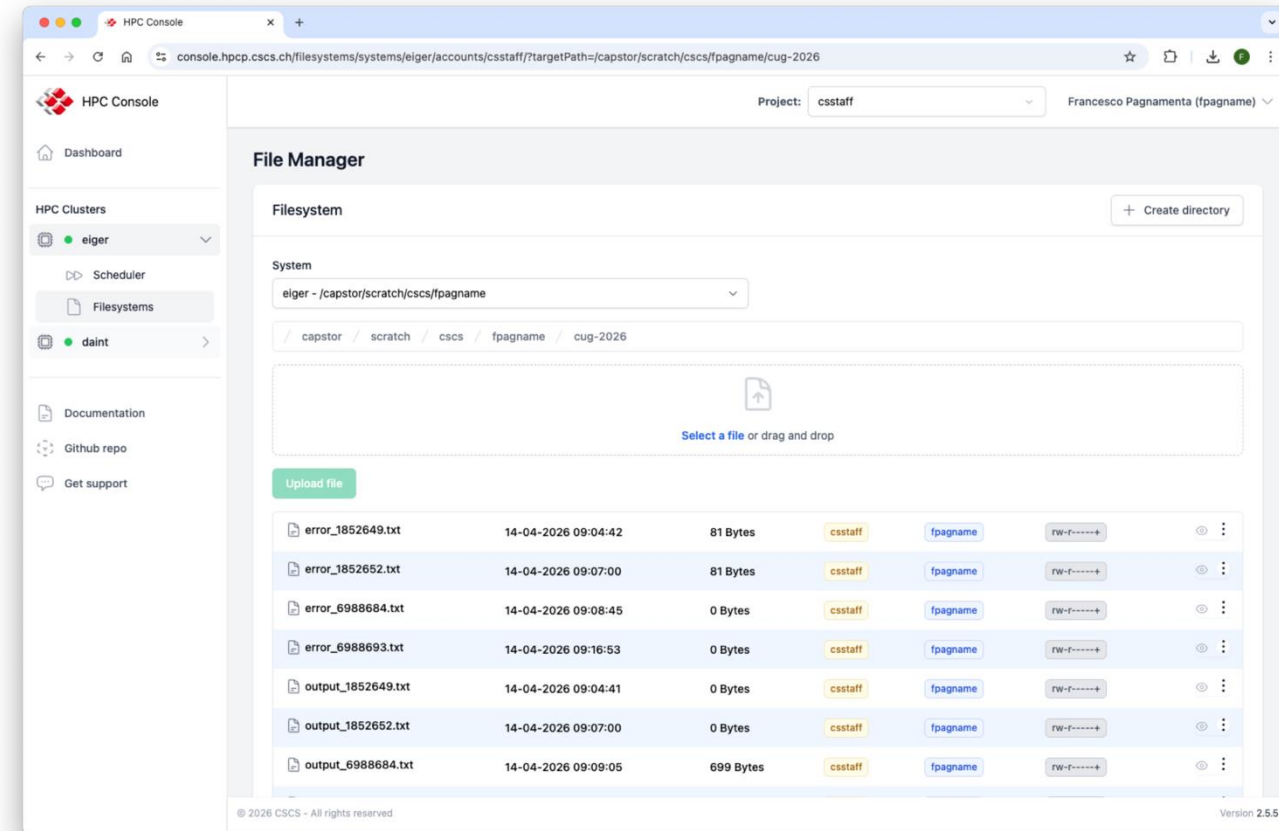
☐ Cluster – Scheduler:

- ☐ List jobs
- ☐ Submit and configure new workloads
- ☐ Monitor running jobs and inspect outputs

☐ Cluster – Filesystem

- ☐ Manage data and perform file operations
- ☐ Preview small files
- ☐ Large data upload

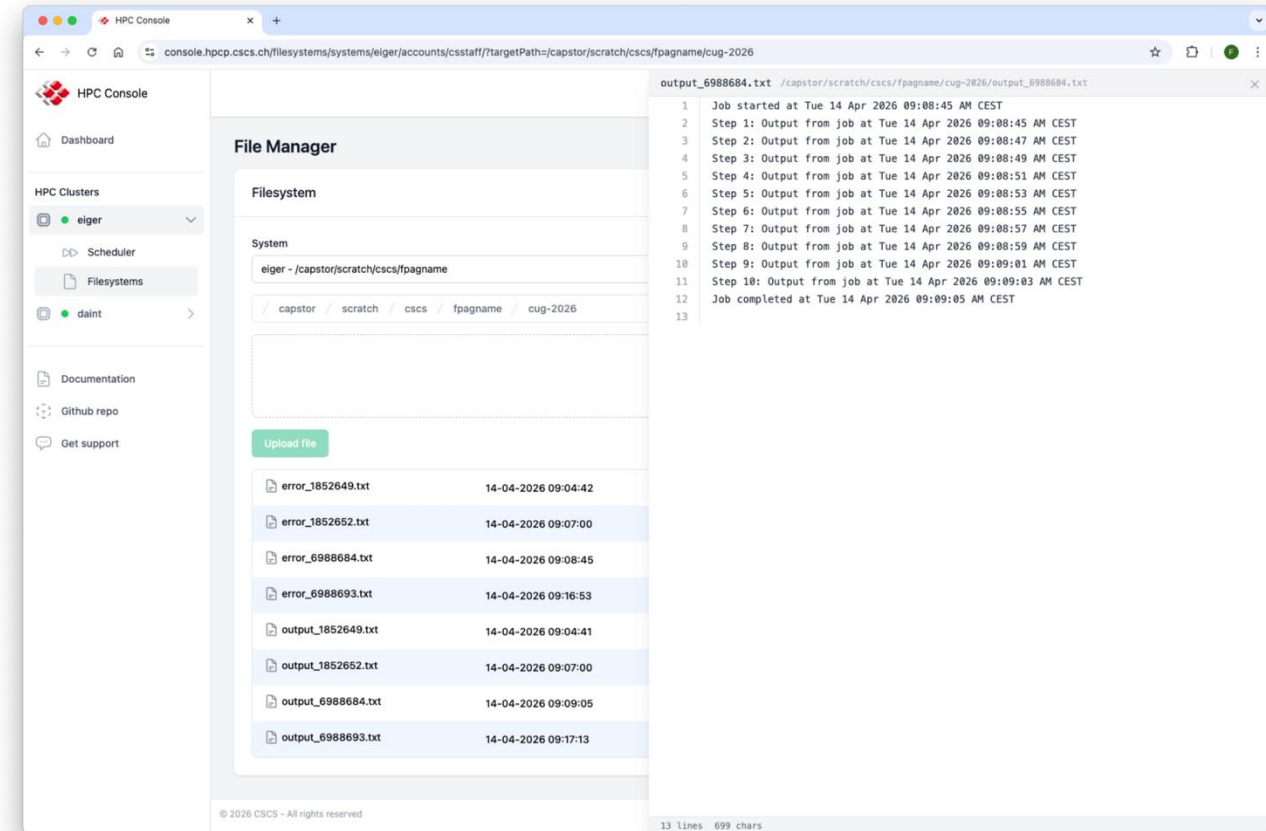
✓ The interface supports common operations on files



Filesystem: data management and preview

- Dashboard:
 - Inspect cluster health and system availability
- Cluster – Scheduler:
 - List jobs
 - Submit and configure new workloads
 - Monitor running jobs and inspect outputs
- Cluster – Filesystem
 - Manage data and perform file operations
 - Preview small files
 - Large data upload

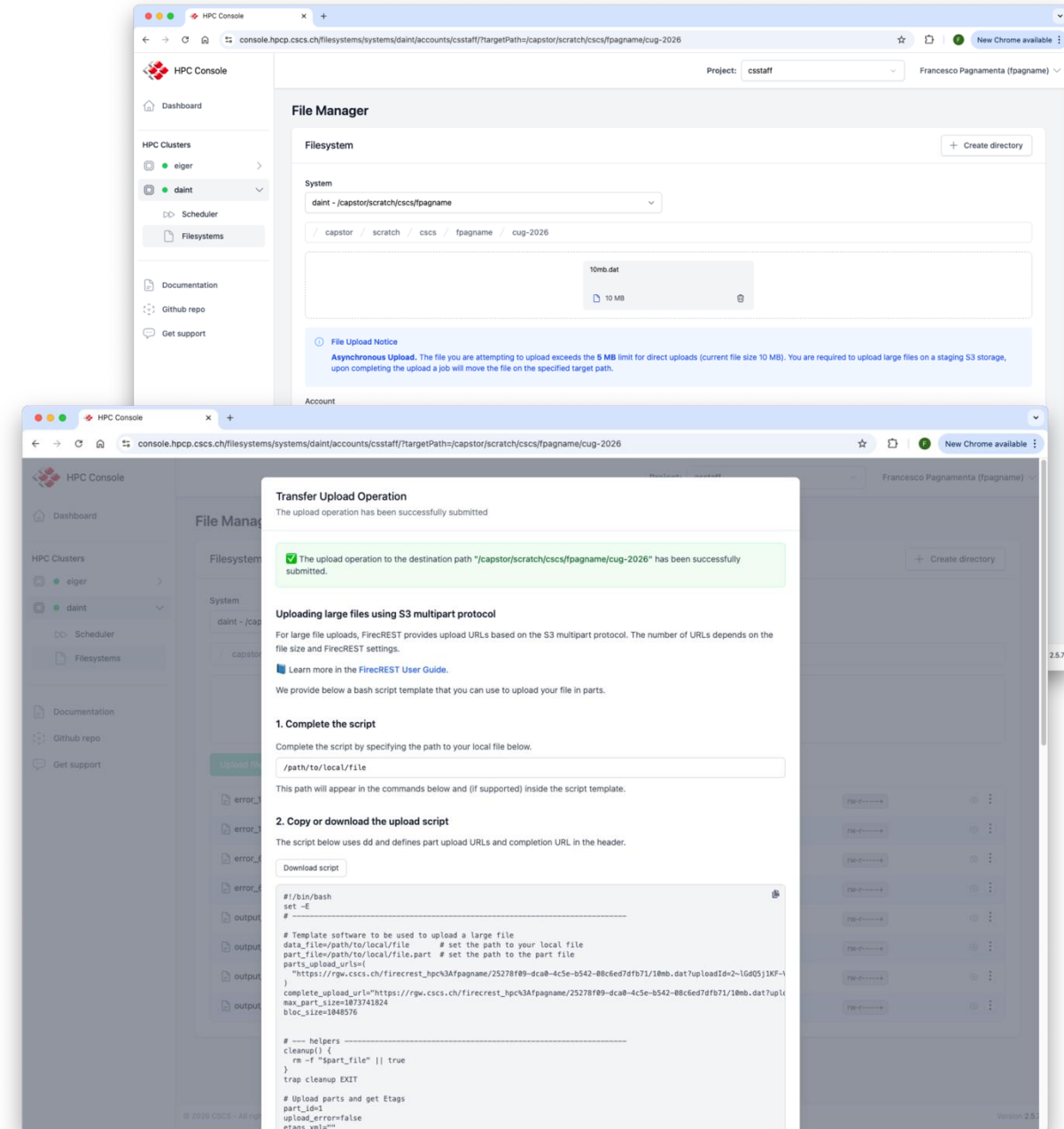
✓ Preview of files under 5MB



Filesystem: data management

- Dashboard:
 - Inspect cluster health and system availability
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✓ For large file uploads, FirecREST support various protocols including S3 (upload URLs based on S3 multipart).

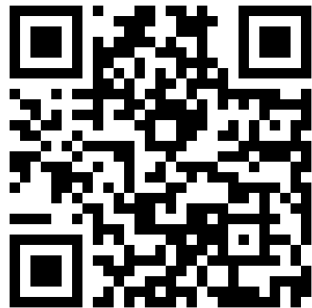


Future work

- Integration with observability tools
 - Grafana dashboards/panels
- Improved performance and scalability
- Whatever you would like to see!

Summary

- The HPC console simplifies common HPC interactions
- It complements existing command-line workflows
- It provides a consistent interface across systems



FirecREST



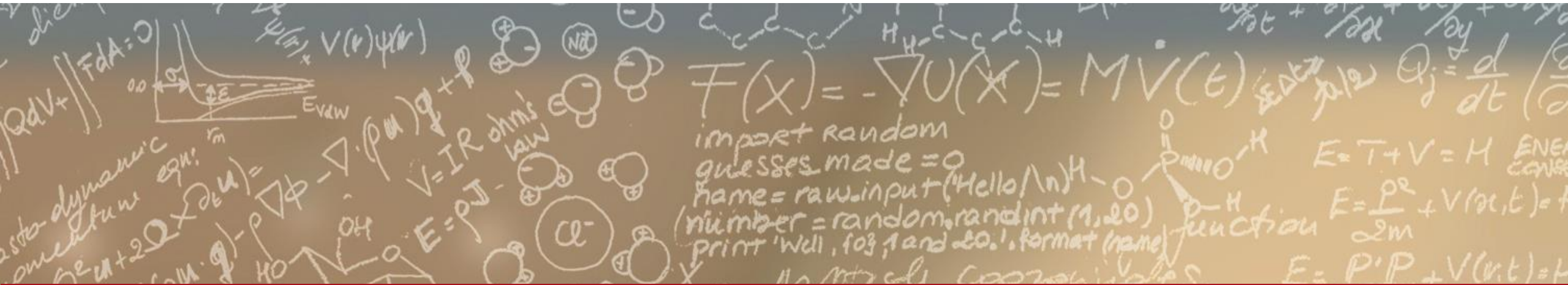
HPC Console



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Thank you.

Demo time and feedback.

Audience feedback

- Were you aware of the HPC Console?
- Have you used it?
- Do you see benefit in it?
- What functionality is most useful?
- What functionality is missing?